

COLE CAMPBELL

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SUMMARY

Data scientist and machine learning engineer, recently graduated cum laude with a B.S. in Data Science from Arizona State University. Experience spans team leadership, statistical modeling, deep learning, and applied AI systems, built and evaluated through projects ranging from academic research to a shipped, paying product.

EDUCATION

Arizona State University · Tempe, AZ

M.S., Data Science, Analytics & Engineering (Computational Mathematics & Data) · *Expected 05/2027*

B.S., Data Science (Computer Science Concentration) · *Completed 05/2026*. GPA 3.53 / 4.00

Barrett, The Honors College · *08/2023 to 05/2026*

TECHNICAL SKILLS

Languages: Python, R, Dart, C/C++, JavaScript (Node.js), SQL, Bash, MATLAB

ML / AI: Deep learning (CNNs, RNNs/LSTMs, Transformers), generative models (Diffusion, GANs, StyleGAN), LLMs & prompt engineering, LoRA/QLoRA/PEFT, transfer learning, neural style transfer, audio/music generation, model evaluation (FID, CLIP, LPIPS, BLEU, perplexity)

Frameworks: PyTorch (torchvision/torchaudio), TensorFlow, Hugging Face (Transformers, Diffusers, Accelerate), scikit-learn, NumPy, Pandas, SciPy, StatsModels, FastAPI, Flutter/Riverpod

Data Science: Regression & statistical modeling, classification, time series, Bayesian, clustering, longitudinal/panel analysis, feature engineering, A/B testing, causal inference, hypothesis testing, visualization (Matplotlib, Seaborn, Plotly)

Data Engineering: ETL & data pipelines, REST API integration, SQLAlchemy + Alembic, PostgreSQL, Firebase, Docker, CUDA, GCP

Tools: Git, Linux, Jupyter, VS Code, OpenCV, librosa, ParaView/VTK, pytest, Weights & Biases, Kaggle/NBA APIs, REST/OpenAPI

Math: Linear algebra, probability, optimization, numerical methods, stochastic processes, statistical theory

PROJECTS

NBA Shot Quality Capstone · Applied Machine Learning & Sports Analytics (2026)

Python · Machine Learning · Statistical Analysis

- Led a four-person team to build an end-to-end machine learning pipeline analyzing 2.9 million NBA shot attempts (2013 to 2025) to separate shot-making skill from shot selection.
- Found that shot-quality-adjusted player rankings diverge from traditional metrics (Spearman's $\rho = 0.39$ vs. FG%), with top performers shifting up to 195 ranking spots once shot difficulty is accounted for.

OfferReady · AI Career Kit (2026)

Gemini API · InstantID · Claude API · DreamBooth LoRA · Stripe

- Designed, built, and shipped an AI career kit end to end, turning a few phone photos into studio-quality headshots, an ATS-approved resume rewrite, LinkedIn updates, and a hosted portfolio.
- Built the generative pipeline on Gemini's image API with a local InstantID + RealVisXL fallback and per-client DreamBooth LoRA fine-tuning, scoring every output against the original for identity consistency; automated the full order pipeline from intake through Stripe payment to delivery.

Multi-Architecture Generative Content Studio · Deep Learning Capstone (2026)

PyTorch · Transformers · Diffusion · GANs · CNNs · RNNs

- Built a unified generative pipeline producing story text, illustrations, character portraits, and music from a single prompt, integrating five architectures (GPT-2, Stable Diffusion, StyleGAN2/3, LSTM) into one system.
- Fine-tuned Stable Diffusion (LoRA + ControlNet) and StyleGAN2 (GAN inversion, latent editing) for controlled scenes and character consistency.
- Benchmarked DCGAN against Stable Diffusion using FID/CLIP metrics; Stable Diffusion had ~390x better image quality, driving the pipeline's generator choice.

Tempo · AI Scheduling Assistant (2026)

FastAPI · Flutter · PostgreSQL · LLM / NLP · Constraint Optimization

- Built a cross-platform AI scheduling app (Flutter, FastAPI) with a constraint-based engine and a natural-language "Ask Tempo" interface that generates, compares, and applies alternative day/week schedule plans.
- Backed by SQLAlchemy + Alembic schema management and Firebase auth; hardened with 360+ backend tests at ~84% coverage.

Portfolio & demos: cjcamp16.github.io

EXPERIENCE

Spinato's Pizzeria · Tempe, AZ

Chef · 01/2024 to 04/2026

- Cross-trained across every role in the kitchen within the first six months, prepping and cooking to order, and showcasing consistency, accuracy, and eagerness to learn.